1. USABILITY ENGINEERING

1.1. Introduction to pragmatic process-oriented design theory for usability engineering usability evaluation

Jakob Nielsen's (1993) publication *Usability Engineering* introduced heuristic evaluation, a method that has withstood the test of time and continues to be widely used as a primary approach for improving user interface design. The well-known set of ten usability heuristics was developed and formalized after the publication of *Usability Engineering* and was not presented as a defined list in the original book. Subsequently, a more formalized definition of *usability engineering* was provided by the International Organization for Standardization in ISO 9241-11:1998, which defines usability as:

> **"the extent to which a system, product, or service can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use."**

Although *usability engineering* had been widely addressed in the literature before Jakob Nielsen's book and has since been further elaborated through numerous usability-related standards, these two cornerstones can be regarded as forming a solid foundation for the *usability engineering* discipline.

The ISO 9241-11:1998 formulation reflects a product-centered and task-oriented perspective, rooted in human–computer interaction (HCI) and workplace system design. The revised ISO 9241-11:2018 retains the same core dimensions but explicitly broadens the definition to include *systems, products and services*, thereby extending its applicability beyond traditional software and hardware systems. In addition, the 2018 standard situates usability more clearly within a holistic understanding of user experience, emphasizing contextual factors. As a result, usability is no longer framed solely as a performance-related quality attribute like in the past when it was just one factor of FURPS model, but rather as a contextual concept relevant to more modern digital services and embedded systems where UX has a bigger role.

Under the revised ISO 9241-11:2018, the concept of user experience includes the following: "user's perceptions and responses that result from the use and/or anticipated use of a system, product or service. Users' perceptions and responses include the users' emotions, beliefs, preferences, perceptions, comfort, behaviours, and accomplishments that occur before, during and after use. User experience is a consequence of brand image, presentation, functionality, system performance, interactive behaviour, and assistive capabilities of a system, product or service. It also results from the user's internal and physical state resulting from prior experiences, attitudes, skills, abilities and personality; and from the *context of use*". The term "*user experience*" can also be used "to refer to competence or processes such as user experience professional, user experience design, user experience method, user experience evaluation, user experience research, user experience department. Human-centred design can only manage those aspects of user experience that result from designed aspects of the interactive system".

Usability evaluation is also one of the core concepts in the *usability engineering* discipline. Nowadays, usability evaluation is an umbrella term that encompasses multiple methods, including usability testing. Usability evaluation is a step-by-step process in which usability-related weaknesses are first identified, then visually documented and finally addressed. In many cases, usability testing begins with qualitative methods such as the think-aloud protocol and may later be supplemented with quantitative techniques, including A/B testing (Viitanen 2023).

Usability evaluation aims to assess how effectively, efficiently and satisfactorily a system under test supports end-users in achieving their primary goals. It is method-oriented and outcome-focused, relying on empirical user data or expert inspection to identify concrete usability problems and to evaluate the quality of the interaction. In contrast, analytical approaches based on the PACT (HCI) model (Weyers et al. 2017), Norman's Human Action Cycle (Norman 2013) and the ACM Curricula framework (ACM SIGCHI 1992) are typically model-driven, explanatory and diagnostic in nature and therefore do not necessarily require empirical user testing. Moreover, neither the PACT (HCI) model nor Norman's model provides standardized usability metrics, such as the System Usability Scale (SUS), nor do they define actionable steps within a usability evaluation workflow (Nieminen 2022). For this reason, these conceptual frameworks are not treated as usability evaluation methods in this study.

One relevant point requires further elaboration. Usability testing based on Nielsen's ten heuristics cannot be conducted optimally by a single individual. Therefore, a usability testing protocol is defined and visualized as an activity diagram to illustrate how usability testing can be most appropriately conducted when the objective is to enhance the utility of the system under test. Prior to presenting the activity diagram of the usability testing protocol, recent empirical studies and the key stages of usability testing are reviewed. The usability testing process is subsequently generalized into a basic scenario to support the development of the activity diagram. In this context, the terms *usability evaluation* and *usability testing* are used interchangeably to refer to the same underlying process aimed at increasing user utility, as described in the following section. The primary objective is to enhance utility for end-users by improving the ease of use of the system under test, thereby enabling tasks to be completed more efficiently.

RQ2: How can an activity diagram be constructed as a rigorously grounded design science artifact that formalizes the essential stages of a collaborative usability engineering evaluation process and supports systematic utility improvement, thereby increasing usability in complex systems development contexts?

This research question is explicitly aligned with Hevner et al.'s **Table 4**, which illustrates the design science research guidelines, as it focuses on the construction, relevance, evaluation, contributions, rigor, search process and communicability of a usability evaluation **artifact** intended to enhance user utility. Hevner et al. (2004, p. 80) argue very strongly that "the goal of design science is utility." Considering the unification echoes of economical imperialism, Lehman's Law (1980), the requirement articulated by Hevner et al. (2004) and the overarching objective of the study, which is the enhancement of end-user utility, the formalization of the activity diagram is focused explicitly on usability evaluation.

Through collaborative usability evaluation, usability is intended to be increased in good faith, which in turn is reflected in improved utility for the end-user (in general). The activity diagram is formalized in a manner that emphasizes real collaboration, as usability evaluation is not considered optimal when conducted by a single evaluator based on own experiences. The constructed activity diagram artifact does not always strictly adhere to UML notation but is more detailed than a process flow diagram. Hence, the primary emphasis is placed on representing workflow structures in a logical, step by step manner through highly visual and explanatory visualization. This approach additionally facilitates effective communication of the artefact across diverse stakeholder groups.

1.2. Recent studies with criticism

The basic usability engineering process has been described, for example, by Wixon and Wilson (1997, p. 658), whose process flow diagram begins with the definition of quantitative usability goals, followed by the specification of desired usability levels for each goal.

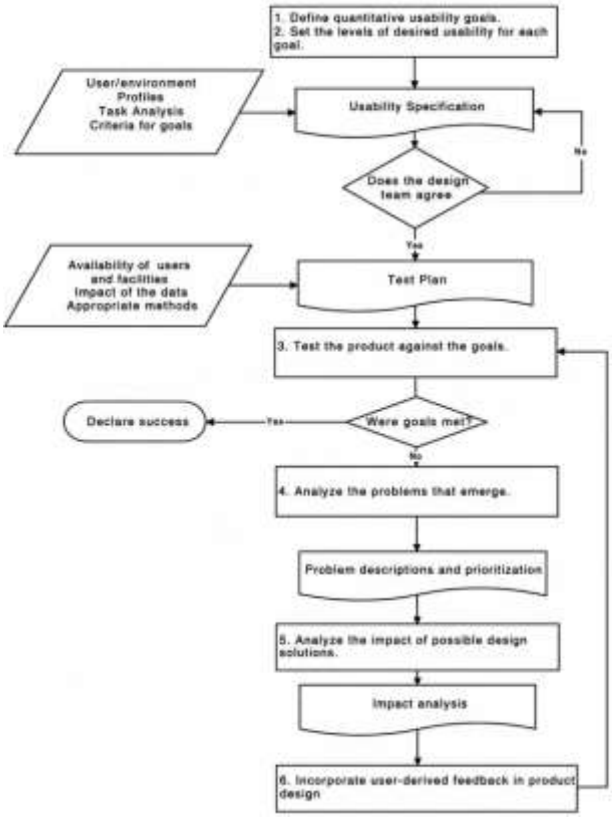


Figure 19. The usability engineering process by Wixon & Wilson (1997).

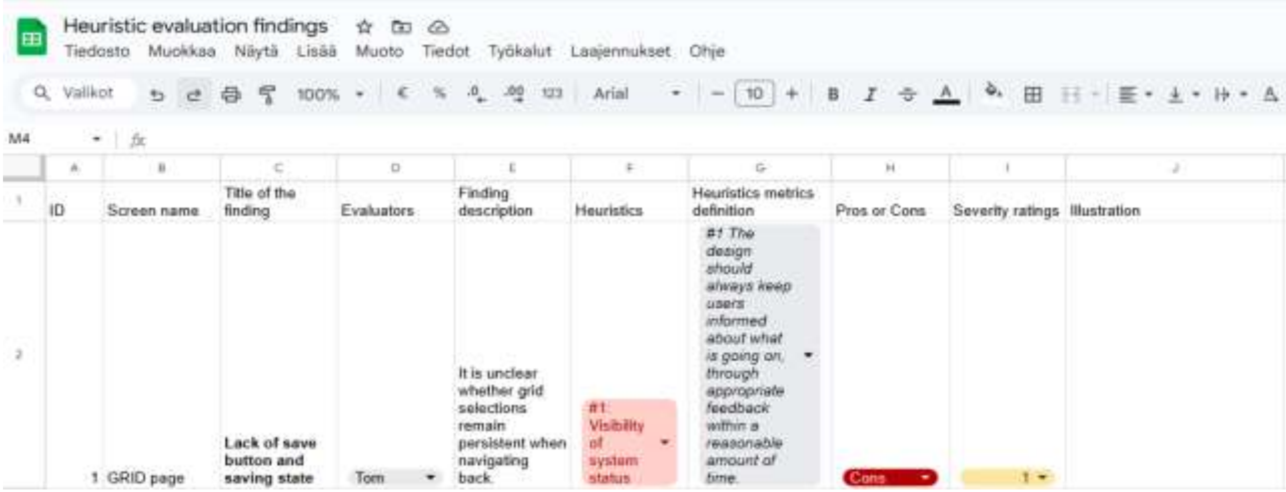
However, as applications have increased substantially in size, particularly in terms of the number of pages they contain, a pragmatic approach is often to prioritize the most significant application pages or routine work flows for usability evaluation, rather than attempting to evaluate all pages individually. Moreover, standardized evaluations such as the System Usability Scale (SUS) can be used to capture overall perceived usability, while evaluators' feedback supports the identification of improvement opportunities and possible later usability improvements. The System Usability Scale (SUS) is a standardized questionnaire-based instrument that provides a reliable measure of overall perceived usability and is commonly used to complement qualitative usability evaluation methods, but for the complex systems evaluation it has its limitations as it is not grounded theoretically.

1.3. Methodology

Wixon and Wilson's (1997, p. 658) process flow diagram begins with the definition of quantitative usability goals, followed by the specification of desired usability levels for each goal. However, the bit contrary methodology for user evaluation adopted in my study is grounded in *HCI theory, Nielsen's Heuristics, ISO standards, chaining cross-LLM prompt template-based experiments, logic, UML based conceptualizations* and *pragmatism*, which means explicitly that my suggestion for the basic workflow and activity diagram begins with the immediate evaluation of notable usability flaws in the selected screens using Nielsen's ten heuristic principles or other relevant ones. To be perfectly clear, the underlying idea of the designed software prototype (later "Contacts") is to improve usability by consolidating most features onto just a single page, where they are presented in a 'Tick-in-the-box Grid'. Personal level, decades of long working experience around the matter as well as the preliminary 50 interviews and peer testers on the software project course already suggest that this has been a successful design choice. This does not imply that the evaluation of the Grid itself constitutes the research problem; rather, it is used solely as an explanatory visual example to illustrate how the activity diagram progresses from one logical step to another. The application of the activity diagrams logic to other software systems might likewise lead to improvements in their levels of usability. All in all, it is likely that the underlying logic is more easily grasped when the pragmatic process-oriented design model is being illustrated using a real-life example, even though the examples would not be production-ready.

Table 7. Heuristic evaluation findings template (Viitanen 2023, modified).

Ticket #	Screen name	Title of the finding	Evaluators	Finding description	Heuristics	Heuristic metrics definition	Pros or cons	Severity ratings	Illustration



I define this stage where the collaborative Team of evaluators gather individual observations at first as pre-analysis in the activity diagram as it forms the first logical steps in categorizing the next workflows. It needs to be highlighted here that once we take e.g. screenshots, then they can be used as multimodal inputs to the LLMs, and the enclosed textual prompt often yield relevant feedback.

The pre-analysis requires the collaborative Team of evaluators to use Google Documentation spreadsheets or similar to create a test report where the columns include the following: ID, Screen name, finding title, evaluators name, finding description, heuristics, heuristics metrics definition, Pros or Cons, severity ratings and illustration. All these columns form a heuristic evaluation findings template. The methods themselves in the ‘Heuristic evaluation findings template’ are based on Jacob Nielsen’s 10 Usability heuristics (1994) and he points out that they are called ‘heuristics’ simply because they are to be seen as broad rules of thumb and not as specific usability guidelines. As the previous page **Table 7**., heuristic evaluation findings template (Viitanen 2023, modified) shows only one of the Nielsen’s heuristics, the below **Table 8** lists them all:

Table 8. Ten usability heuristics (Nielsen, 1994).

#	**Nielsens heuristic**	**Brief description**
i.)	Visibility of the system status	The design should always keep users informed about what is going on, through appropriate feedback within a reasonable amount of time.
ii.)	Match between system and the real world	The design should speak the users’ language. Use words, phrases and concepts familiar to the user, rather than internal jargon. Follow real-world conventions, making information appear in a natural and logical order.
iii.)	User control and freedom	Users often perform actions by mistake. They need a clearly marked “emergency exit” to leave the unwanted action without having to go through an extended process.
iv.)	Consistency and standards	Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform and industry conventions.
v.)	Error prevention	Good error messages are important, but the best designs carefully prevent problems occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.
vi.)	Recognition rather than recall	Minimize the user’s memory load by making elements, actions, and options visible. The user should not have to remember information from one part of the interface to another. Information required to use the design (e.g field labels or menu items) should be visible or easily retrievable when needed.
vii.)	Flexibility and efficiency of use	Shortcuts – hidden from novice users – may speed up the interaction for the expert user so that the design can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.
viii.)	Aesthetic and minimalistic design	Interfaces should not contain information that is irrelevant or rarely needed. Every extra unit of information in an interface competes with the relevant units of information and diminishes their relative visibility.

ix.)	Help users recognize, diagnose, and recover from errors:	Error messages should be expressed in plain language (no error codes), precisely indicate the problem and constructively suggest a solution. These error messages should also be presented with visual treatments that will help users notice and recognize them.
x.)	Help and documentation:	it's best if the system doesn't need any additional explanation. However, it may be necessary to provide documentation to help users understand how to complete their tasks.

The methods will reveal usability related problems and as they differ from each other, then it is good to have a category in place which categorizes such problems in relation to their level of severity. The severity ratings can be as follows:

Table 9. Severity scale for collaborative evaluations in the context of usability engineering evaluation process.

SCALE	Brief description
0	This is not a usability problem at all and does not really affect tasks to perform as planned.
1	Cosmetic problem. Need not be fixed unless extra time is available on project. This represents minimal impact on usability.
2	Minor usability problem. Fixing this should be given low priority.
3	Major usability problem. Important to fix when the tasks do not perform as users want and reduce satisfaction, so should be given high priority.
4	Usability catastrophe. It is imperative to fix this before product can be released as the task completion cannot be guaranteed.

Standardizing a consistent workflow for all usability evaluators in the collaborative team is a sound methodological practice, as it helps ensure that the evaluation scale remains similar across evaluators. The objective of this phase is to facilitate subsequent aggregation by ensuring that all collaborators follow similar workflow rules and that decisions are, to some extent, based on consensus. The usability engineering usability evaluation process may also incorporate additional methods to maintain a standardized workflow, although the following ones listed seemed to be enough in one real-life Case Analysis led by Viitanen (2023) and I have extended this method list by bringing the GenAI viewpoints in the form of cross-LLM prompt template that can be chained across alternative LLMs. The usability evaluations process flow methods can be now defined as follows:

Table 10. Additional methods to standardize the usability evaluation workflow.

Method	**Brief description**
Independence	The screen views are examined independently by each evaluator, with a minimum of two review iterations performed before completion of the final test report.
GenAI	Cross-LLM prompt template chained
Aggregation	The usability evaluation outcomes produced by all evaluators are combined into a single report. The usability evaluation findings from all evaluators can be compiled into a unified document.
Consolidation	Overlapping findings identified by multiple evaluators may be merged.
Catastrophe consensus	Findings classified as usability catastrophes are subject to a consensus decision to validate this severity rating.
Consensus for major usability problem	Tickets assigned a severe rating are illustrated in a manner that reflects their severity.
Recommend solutions.	Addressing identified improvement opportunities.
Reporting	Synthetizing the found usability observations

The focus of the evaluation is placed typically on workflows rather than on isolated user interface elements or individual screen level features, which should be defined clearly so that the relevant stakeholder for each workflow is self-explanatory. The reporting of results is categorized into positive and negative findings, where positive findings refer to existing effective implementations and their potential for further improvement, while negative findings address identified usability flaws of varying severity. As the objective is the improvement of utility, the specific source of the improvement is considered secondary, since increased utility may result either from the elimination of usability flaws or through feature improvements. The final report should include a summary of identified positive findings in order to preserve effective implementations and to identify these areas of excellence in which existing strengths may be further improved. This approach is intended also to support a polite and constructive tone in face-to-face business discussions by providing an opportunity for maintaining positive dialogue. In practice, however, the final report primarily consists of improvement proposals derived from identified usability weaknesses, as modern methods for detecting such weaknesses have become highly effective.

Next a few words about the chosen metric in the activity diagram, the system usability scale (SUS). In the field of HCI and systems engineering, the system usability scale (SUS) is a simple, ten-item attitude *Likert scale* giving a global view of subjective assessments of usability. SUS was developed by Brooke (1996) at Digital Equipment Corporation in the UK as a tool to be used in usability engineering of electronic office systems. According to ISO 9241 Part 11, system usability can only be measured by considering the *context of use* including the users, their purposes and the environment, with measurements encompassing effectiveness, efficiency and satisfaction aspects. The SUS is effective, because its simplistic 0-100 score enables easy usability comparisons across even different types of systems. However, this simplified approach serves as both its strength and also limitation on many occasions (Brooke (2013), ISO 9241-11:1998, ISO 9241-11:2018).

The choice of SUS (and later also QQ) as metrics is based on the holistic concept of the PUSA-X framework, which focuses on complex systems rather than traditional HCI UI usability testing. Having a quantifiable metric also enables longitudinal studies in the future, allowing the system to be compared against the competitive solutions, for instance. The empirical research further highlights several relevant additional insights, for instance, Lewis and Sauro (2009, p. 1) suggested a two-factor orthogonal structure, where practitioners can use the existing SUS questionnaire as usual to get the overall usability score, but can also break down the results into two own scores. On the one hand one can measure how usable the system is and on the other hand measure how easy it is to learn. This gives usability evaluators more versatile insights.

> "Factor analysis of two independent SUS data sets reveals that the SUS actually has two factors – Usable (8 items) and Learnable (2 items – specifically, Items 4 and 10). These new scales have reasonable reliability (coefficient alpha of .91 and .70, respectively). They correlate highly with the overall SUS (r = .985 and .784, respectively) and correlate significantly with one another (r = .664), but at a low enough level to use as separate scales. A sensitivity analysis using data from 19 tests had a significant Test by Scale interaction, providing additional evidence of the differential utility of the new scales. Practitioners can continue to use the current SUS as is, but, at no extra cost, can also take advantage of these new scales to extract additional information from their SUS data."

It could naturally be asked why such an old metric has been chosen, one that was invented before modern websites, let alone smart devices had even arrived. I quote the creator Brooke (2013) in his article 'SUS a retrospective':

> "SUS has proved its value over the years. The efforts of all those people who picked it up and did all of their work quite independently of me have provided additional evidence that it's a tool worth using and probably has many more years of valuable life left in it."

The original paper on SUS by Brooke (1996) continues to be cited regularly across HCI, UX research and software engineering literature, with a citation frequency that is rarely achieved by researchers over the course of an entire career, proving its impact as a reliable, low-cost usability scale that can be used for global assessments of systems usability.

1.4. Results

1.4.1. The pre-analysis

First of all, the pre-analysis stage forms the beginning steps in the usability engineering process being modelled. But firstly, for the sake of clarification, it should be noted that the pre-analysis phase could alternatively begin with the definition of so-called *persona* concepts, which typically refer to fictitious or real end users in the early stage of software project design and modelling. However, in most cases, the end users, user groups and corresponding use cases to be served by the application or complex system are already clearly identified by the project leaders. Consequently, the definition of *personas* concepts may be considered largely irrelevant to the overall pre-analysis level usability engineering evaluation process now.

The *Contacts* prototype application can now be used to test the initial workflows of the activity diagram. The illustration column presents the Grid, which is the core page of this particular application and this preliminary analysis identifies a potential usability weakness related to changes in system state, meaning the #1: visibility of system status.

Figure 20. ‘Tick-in-the-box’ GRID.

In the above framework, the relevant columns used to record usability-related observations in software or complex systems are now defined. In contrast to the *Contacts* prototype application, modern digital systems may contain numerous screens with substantially different functionalities. Accordingly, each screen should be assigned a descriptive name to support collaborative analysis. This practice facilitates both collaboration and traceability, as a single screen may easily generate more than one hundred usability observations according to my personal experiences in the software development industry. To be more pragmatic later on, the usability engineering evaluation process should focus on workflows (e.g., *User Journey Map*) also and these workflows should be defined in a manner that clearly identifies the stakeholder involved. In other words, following UML notation for a User Group Actor and a Use Case together e.g. with an activity or sequence diagram once the screens have been refactored.

The pre-analysis phase includes also some basic measurements in the context of complex smart glass systems development. The basic measurements are discussed in the next chapter and these include the CPU utilization % (zones) factors and inductive charging viewpoints after weighing what are most relevant. The various stakeholders do not typically know what goes under the hood either at the source code or physical hardware level, but they can naturally sense when the overall usability feels suboptimal, for example due to latency. Hence, a process-oriented design modeling for improved usability should be capable in aiding to measure the different source code or physical hardware components causing dissatisfaction. As such, now pre-analysis should also determine the level of safety for the CPU utilization and inductive charging time for the electrical gadget. In some cases, also strength of the hardware components physical parts.

1.4.2. Independent evaluation

The second phase in the activity diagram is to group the observations in relation to their level of severity, so that one can start implementing solutions especially to the ones labeled as usability catastrophes which is the 4th option in the scale. The scale for this can be seen below with a description. The methods will reveal usability related problems and as they differ from each other, then it is good to have a category in place which categorizes such problems in relation to their level of severity. The severity ratings can be as follows:

Table 11. Severity scale for collaborative evaluations in the context of usability engineering evaluation process.

SCALE	DESCRIPTION
0	This is not a usability problem at all and does not really affect tasks to perform as planned.
1	Cosmetic problem. Need not be fixed unless extra time is available on project. This represents minimal impact on usability.
2	Minor usability problem. Fixing this should be given low priority.
3	Major usability problem. Important to fix when the tasks do not perform as users want and reduces satisfaction, so should be given high priority.

4	Usability catastrophe. Imperative to fix this before product can be released as the task completion cannot be guaranteed.

Now that we have also the heuristics, the CPU utilization % (zones) factor and inductive charging under evaluation, then also feedback from this complex systems contexts effectivity is wanted.

1.4.3. Chaining cross-LLM prompt template

The created cross-LLM prompt template was already thoroughly discussed in the Prompt Engineering chapter and its efficiency was illustrated. Within the description of this *process-oriented design model* for usability engineering, it needs to be pointed out that this workflow is also intended to be independent. As a result, independent researchers are enabled to formulate their own ideas regarding what should be considered the most appropriate LLM outputs and to discuss, during the aggregation phase, why especially their findings should be regarded as the most optimal ones for system improvement purposes. At a general level, obtaining efficient outputs step by step from the LLMs represents one aspect of the process, while convincing other collaborating team members represents truly another, in a manner comparable to independent heuristic evaluation.

When Nielsen's heuristic-based usability evaluation is reconsidered, it can be observed that it is highly effective in generating ideas for improving interface elements such as landing pages or the ways in which administrative users record different types of data. However, reliance on heuristics alone is no longer required, as the PUSA-X framework enables the inclusion of *multimodal* inputs and the generation of outputs addressing a wide range of potential improvement areas. As a result, the complex system under test can be evaluated in a more holistic manner. In other words, the usability engineering evaluation process is transformed into a holistic approach that evaluates complex systems, such as smart glasses, within more relevant usage contexts. Hence, a focus on usability improvements limited to a narrow context can be regarded as somewhat outdated. Fortunately, this approach enables the needs of various stakeholders to be analyzed and served more effectively. However, the GenAI causes challenges too, because chaining *cross-LLM prompt template* results easily hundreds, sometimes even thousands of rational suggestions. Who is to say which are the optimal ones?

Next, an illustration is provided showing how the *cross-LLM prompt template* processes multimodal input and subsequently produces outputs that offer novel ideas to be filtered and prioritized prior to aggregation. In this context, a holistic approach implies that the evaluation is no longer narrowly focused on UI or button-level improvements. Instead, the prompt template supports a holistic application and rather than listing all possible use cases here, it is more worthwhile to note that there are few areas within complex systems engineering in which the generated outputs would be of limited or no practical value.

<table>
<tr><th></th><th>WRITE PROMPT</th><th>ADD TO THE LLM</th></tr>
<tr><td>DEFINE TASK
Give me …
Tell me …
Translate:
Remove grammar errors:</td><td></td><td rowspan="6">[TASK]

[EXAMPLAR]

[OUTPUT FORMAT]

[STYLE]

[ROLE]

[ADDITIONAL INFORMATION]</td></tr>
<tr><td>ADD EXAMPLAR</td><td></td></tr>
<tr><td>OUTPUT FORMAT
<table>
<tr><td>Text</td><td>Audio</td><td>Video</td></tr>
<tr><td>TXT
PDF
JSON
XML</td><td>MP3
Wav</td><td>MP4</td></tr>
<tr><td>Images</td><td>3D</td><td>Data</td></tr>
<tr><td>JPG
JPEG
PNG
GIF</td><td>STL
OBJ
FBX</td><td>CSV
XLSX
SQL dump</td></tr>
</table></td><td></td></tr>
<tr><td>STYLE</td><td></td></tr>
<tr><td>ROLE</td><td></td></tr>
<tr><td>ADDITIONAL INFORMATION</td><td></td></tr>
</table>

Figure 21. Standardized multimodal cross-LLM prompt template for complex systems analysis.

The prompt template shown above includes a section titled *"Add to the LLM"* in the far-right column. This template is designed to function as a reminder of the information that needs to be provided as input to the various LLMs. For example, if the objective is to query potential weaknesses in specific application screens, the required commands can be formulated by following the prompt template and simply enclosing an additional screenshot as additional information. No further input is required. Leading LLMs naturally provide their own recommended prompt engineering frameworks, such as the CRISPE prompt engineering framework for ChatGPT. These frameworks, like the field of prompt engineering itself, are constantly undergoing rapid changes and iterative improvements. Even many journals articles these days are '*dead on arrival*' when finally getting published. Therefore, identifying a framework that best fits one's own specific needs may be the most effective way to learn this field, particularly through active practical testing and evaluation of the available frameworks. Also, it is a good point to keep in mind, that these days the LLMs are becoming focused more and more, so paying attention to what niches are available is recommended. See the reference for further details (Aixploria 2026).

In this next example, the principal conceptual instruments are the *cross-LLM prompt template* and *prompt chaining*. At this stage, a transition from the analytical phase to the planning and execution phase is visualized. The planning phase involves completing the *prompt template* with the required elements and subsequently using the resulting outputs as inputs for another base LLM model. Presented below is the original starting *prompt* used for this analysis, which is based on my standardized multimodal cross-LLM *prompt template* for complex systems analysis.

[TASK] Evaluate the provided user interface screenshot image and identify usability improvement ideas ensuring that all findings are explicitly categorized according to Nielsen's 10 usability heuristics. [EXAMPLAR] Provide a categorized list using Nielsen's usability heuristics as the categorization structure, where each entry includes the heuristic name, a short and concrete usability issue or improvement idea, and single-sentence explanation, and include only those heuristics for which relevant findings exist. [OUTPUT FORMAT] Categorized list. [STYLE] Professional. [ROLE] I am an expert in usability engineering and user interface design with deep knowledge of Nielsen's usability heuristics. [ADDITIONAL INFORMATION] A screenshot of the existing tick-in-the-box grid design of the user interface can be considered as an input.

Figure 22. Original starting prompt.

The first tactic involved applying the same prompt template to all six selected top LLMs. As the models performed as anticipated and produced broadly similar outputs, it became evident that the LLMs should be utilized in a more differentiated manner. This approach enables the full benefits of each model to be realized while avoiding unnecessary repetitive effort, consistent with the DRY (*Don't Repeat Yourself*) principle taught even in introductory programming courses. Accordingly, an additional instruction was incorporated into the [ADDITIONAL INFORMATION] section: "Remove repeated points and add only new observations for the heuristics that are missing, so the output includes only fresh insights not already listed."

Therefore, this second tactic was selected for testing and further elaboration. The continuing prompt used for this analysis, which is based on my standardized multimodal cross-LLM prompt template for complex systems analysis, is defined as follows:

[TASK] Evaluate the provided user interface screenshot image and identify usability improvement ideas ensuring that all findings are explicitly categorized according to Nielsen's 10 usability heuristics. [EXAMPLAR] Provide a categorized list using Nielsen's usability heuristics as the categorization structure, where each entry includes the heuristic name, a short and concrete usability issue or improvement idea, and a single-sentence explanation, and include only those heuristics for which relevant findings exist. [OUTPUT FORMAT] Categorized list. [STYLE] Professional. [ROLE] I am an expert in usability engineering and user interface design with deep knowledge of Nielsen's usability heuristics. [ADDITIONAL INFORMATION] Remove repeated points and add only new observations for the heuristics that are missing, so the output includes only fresh insights not already listed. A screenshot of the existing tick-in-the-box grid design of the user interface can be considered as an input:

Figure 23. Revised prompt based on DRY design principles.

AIXPLORIA is an online service that provides accurate information about leading AI tools for various use cases and purposes. Six tools were selected for this study, and the context window was a critical design consideration, as it was necessary to be able to download all previous outputs to ensure that earlier observations would not be repeated. One relevant point worth mentioning here is also whether there is going to be a context engineering discipline in the future. We just have to wait and see.

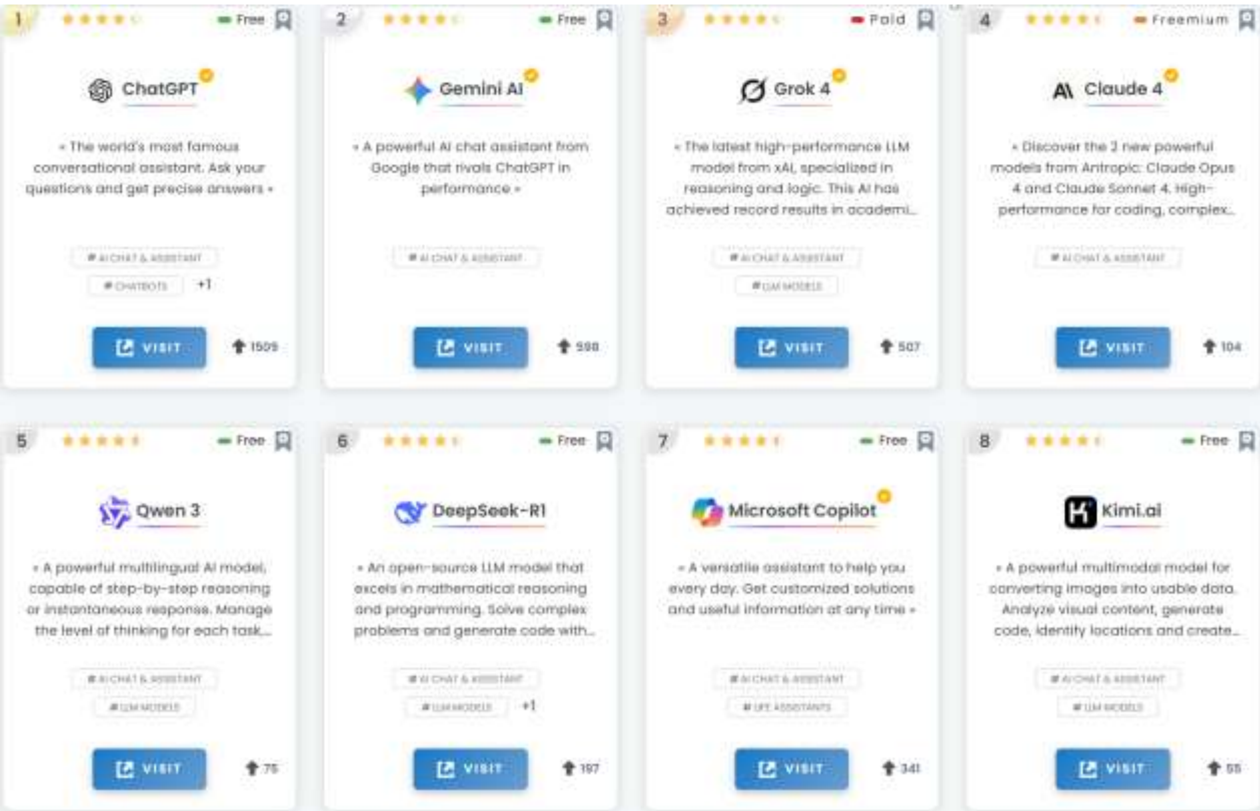


Figure 24. Top-Ranked AI Chat and Assistant Tools Identified by AIXPLORIA (14 February 2026).

Table 12. Prompt template model illustration for seeking a second opinion.

Prompt Stage	**ChatGPT**	**Claude**	**Gemini AI**	**Qwen3-Max**	**KIMI K2.5**	**DeepSeek-V3.2**
Original prompt	1st try	2nd try + [Additional Information]	3rd try + [Additional Information]	4rd try + [Additional Information]	5th try + [Additional Information]	6th try + [Additional Information]
Result 1	"A"					
Result 2		"B"				
Result 3			"C"			
Result 4				"D"		
Result 5					"E"	
Result 6						"F"
Total number of suggestions						**SUM:**
Interpretation of the outputs	Success	Success / Not as planned	Success / Not as planned	Not as planned (Repeats Gemini)	Success / Not as planned	Success / Not as planned

The initial set of outputs from the ChatGPT was sufficiently convincing to demonstrate the effectiveness of the underlying methodological approach. Starting from the Claude phase, an additional prompt element labeled [ADDITIONAL INFORMATION] was revised, instructing the model to "Remove repeated points and add only new observations for the heuristics that are missing, so the output includes only fresh insights not already listed." This element was applied to a query containing output generated earlier by ChatGPT. This practice was repeated in all steps.

The resulting outcome indicates that the proposed prompt template pattern functions as intended within the examined context, yielding a substantially expanded set of heuristic observations. Due to the extensive volume of the generated output, the complete results are provided in this study as additional Appendices. Overall, the findings demonstrate that the prompt template performs according to its design objectives.

The high number of observations produced for a single screen is considerable and necessitates a subsequent aggregation phase. However, individual usability evaluators are expected to review, filter and prioritize their observations independently prior to aggregation. The subsequent aggregation phase is intended to be conducted collaboratively, enabling a more systematic review, filtering, prioritization and selection of the most effective usability improvement proposals for implementation.

One additional methodological consideration should also be noted. If the prompt template is applied without explicitly referencing Nielsen's heuristic framework, the volume of generated results increases further. This alternative approach is not examined in greater detail here, in order to remain within the defined practical scope of the research. Furthermore, the proposed prompt template is also applicable in other multimodal contexts. While its use could provide additional insights from hardware-oriented perspectives, this possibility is not examined in further detail in the present study, in order to remain within the defined research scope.

1.4.4. Aggregation

Now is the time for the collaborative team to start discussing about their observations from the individual *inspection-based evaluation* referring to conducted individual analyses, so that the team can reach a consensus what are the most severe catastrophe level implementations so that they can be solved (Viitanen 2023). I am not going to define the term *aggregation* to the point of preciseness here, but please do note that it does not have the UML context meaning. Now, the collaborative team needs to reach a consensus by aggregating the observations so they become unified themes and on occasion an affinity diagram can help a lot because it enables them to see everything nicely at once. The key observation categories are usability flaws and good features already in place which can sound a bit like the SWOT analysis but one can leave the opportunities and threats categories out here. Furthermore, one should not merely describe the observations but be a lot more analytical especially in the context of good features already in place because often good features can be made even better so this serves as a reminder that usability improvements happen in my personal view by getting rid of usability flaws but more importantly also by improving the good already existing implementations. One could ask perhaps in the context of weak implementation a question: "How can this be solved?" and in the context of good implementation "What additional value could an improved feature bring to a certain user group?". Lastly, as a reminder, aggregation attempts to reach a consensus, so that the usability engineering process evaluation can proceed to the next stage. Reaching a consensus is of course not always a simple and straightforward negotiation, as often the teams involve participants whose opinions have more weight due to a position or prior experience and so on.

1.4.5. Rapid assessment process

At this stage the time has come to interview various user group representatives in such a manner that they will discuss the functionality of the most relevant features for certain use cases in a system under test (SUT). The qualitative research methodology literature defines many possible alternatives for conducting this phase and I have chosen a process by Beebe (2008, p. 726) labeled as the rapid assessment process, RAP:

> "Rapid assessment is defined as intensive, team-based qualitative inquiry using triangulation, iterative data analysis, and additional data collection to quickly develop a preliminary understanding of a situation from the insider's perspective. Rapid assessment allows a team of at least two researchers to investigate complicated situations where issues are not yet well defined and where there is not sufficient time or other resources for long-term, traditional qualitative research."

Beebe (2008) also mentions the BusinessWeek magazine article in June 2006 claiming that "ethnography is the new core competence" as that article described the use of ethnography to "develop a deep understanding of how people live and work" but most researchers do not have enough resources to do it. I agree that being part of a business operation in some division from one week to a couple of months can be a real eye-opener how a certain business operates although I would not call this type of internship work ethnographic method although it must also be said that this is great example how also the gray literature can contribute to the advancement of research methodology just marvelously. Moreover, now when the various LLMs have provided more time resources elsewhere, then why not use this time to learn what would be the most optimal process automation steps for certain routine tasks by becoming an observing ethnographic to some extent if it makes the requirement specifications correct ones in first try.

The acronym RAP can be described as *semi-structured interview* and *directed conversations* mainly because the rapid assessment is based on talking with people and getting them to tell their side of the story. The objective can be at first a bit surprising as “objective is to communicate with participants using their vocabulary and rhythm and is *not* to get answers to questions” but to talk freely and frankly. The directed group discussions “involve the entire team talking with one or more local participants”, but this is *not* sequential interviewing by individual members of the team. Furthermore, individuals with whom the RAP team talks are purposefully selected, and they are not a sample, but “they are selected not because they are believed to be average but rather because they are believed to represent the diversity found in the local situation”. The last hints in this respect by Beebe (2008, p. 727) is so that “the RAP team should seek out the poorer, less articulate, more upset individuals and those least like the members of the RAP team”. Also, the *semi-structured interviewing* and *directed conversations* need to be organized in such a manner for participants that it allows the observations of team interaction and listening of the participants for the purpose of additional data collection.

One approach to further accelerating the Rapid Assessment Process is the use of online *semi-structured interviews* conducted via video conferencing applications for recording purposes, although a substantial portion of the interactions within the participant group may remain unnoticed. Nevertheless, the use of *semi-structured interviews and directed conversations* logically advances the usability engineering evaluation process to the next stage once the context has been sufficiently familiarized. At this point, the planning and conduct of *cognitive interviews* becomes possible where applicable. Although such *cognitive interviews* may also be conducted online, a significant amount of nonverbal interaction may again be easily overlooked.

1.4.6. Interview guide

According to Given (2008, pp. 469-470) an *interview guide* means a summary of the content like topics and questions that the researchers cover during interviews. An interview guide allows certain flexibility in how the individual interviewee can answer the questions or in which order. At one end of the spectrum, interview guides may offer only minimal guidance, resulting in less structured interviews that focus primarily on eliciting “the participant’s own perspective on the research topic”.

The least structured approaches include methods such as James Spradley's ethnographic interviewing, which sets aside substantively predefined topics and instead relies on broad, open-ended questions to draw out the participant's own accounts. For instance, Spradley's approach typically starts with "grand tour" questions, such as "Tell me about a typical instance of …". These grand tour questions help identify the core elements that the participant associates with the research topic after which the interviewer follows up with "mini-tour" questions like "You mentioned [topic]; could you tell me more about …?". Given (2008, pp. 469-470) also points out that at the opposite end of the spectrum, interview guides may include detailed specifications to ensure that all topics of interest to the researcher are comprehensively addressed, and, "in contrast to unstructured interviews, the most highly structured format is the survey questionnaire", which functions as an interview guide by specifying in advance both the content of the questions and the range of possible responses.

In practice I have observed that many researchers find that qualitative interviews tend to evolve into semi-structured interviews during data collection. In collaborative research teams, some team members may ask questions in a more conversational manner, while others adhere closely to the predetermined interview guide based questions. This situation is common when multiple researchers are involved in conducting interviews and it is not necessarily problematic, as rich and valuable data can still be collected. During the transcription and recommendation phase, a larger sample size can help compensate for instances in which certain pieces of information may be missing from individual interviews. One way to maintain shared understanding and peace of mind within a collaborative research team is to explicitly define the interview guide as a general framework rather than a rigid script or the other way around depending on each context. It is also helpful to agree in advance that flexibility is permitted in how questions are asked.

Morgan&Guevara (2008) point out that "one likely reason for the popularity of question-based interview guides is their ability to suggest probes and follow-up questions that elaborate on the basic set of questions". However, they also note that general probes such as "Can you give me an example?" and "What else can you tell me about that?" are rarely written out explicitly in interview guides. I recommend including such follow-up prompts in the interview guide document, as the goal is to elicit rich and detailed data from end users, who are often the true experts on how a system works or should work in practice. Remaining silent during interviews can limit the depth of insight gained and hinder opportunities for system improvement.

1.4.7. Cognitive interview

According to Given (2008, pp. 89-90) *cognitive interviewing* can be traced back to late 1970s when it was developed through the interdisciplinary efforts by survey design methodologists and cognitive psychologists and is increasingly used "in the evaluation of technology interfaces such as websites and tools for informatics" In addition, a similar *cognitive walkthrough* method is also used in marketing to improve the evaluation of products (Wharton et al. 1993, pp. 1-4). In this methodology area, important contributions have been made by cognitive interviewing to both instrument development and survey design. Furthermore, according to Given (2008, pp. 89-91) the two main approaches used in cognitive interviewing are ***verbal probing*** and ***think-aloud* techniques**. *Verbal probing* is conducted after respondents have completed the survey questions at which point additional questions are asked to explore how the questions were understood. Through this process, the potential problems can be identified well in advance. Also, relevant corrections can be made to the study design whenever problems emerge. In *think-aloud* interviewing, the respondent is asked to verbalize the thoughts that come to mind while simultaneously answering questions or completing tasks within an interface. This approach allows the interviewee's responses and opinions to be recorded for later analysis.

According to Knafl (2008, p. 91) substantial improvements in instrument quality related to study design can be achieved by applying cognitive interviewing. The concept of *instrument quality* may lead into some interpretation difficulties but in this study, I define the instrument quality simply as the extent to which a research questionnaire accurately measures the responses.

Based on own experience, the use of *verbal probing* has been found to effectively support the rapid reformulation of research questionnaire items whenever needed, thus improving their clarity and ease of understanding. This in turn helps the construction of clearer responses by interviewees. Consequently, the conduct of a limited number of pilot interviews is considered a pragmatic approach, as this practice enables the revision of questionnaire items in cases where interviewees experience difficulties in responding and this practice is recommended also by Viitanen (2023). Typically just one pilot interview does the trick. As a result, now *think-aloud techniques* can be applied with fewer difficulties once research questions have been clearly formulated and are easily understood, allowing interviewees to express their opinions with greater ease. Furthermore, in situations where interviewees still continue to experience difficulties with certain usability survey questions, additional probing may be used, such as asking respondents to 'comment freely on any additional thoughts that come to mind'. Overall, qualitative research need not involve overly complex methodological procedures, as relevant and well-formulated questions posed to real end users can easily reveal current needs, successful features already in place and concerns.

User groups may be observed at both the individual interview level and in larger group interview settings, with participants being informed in advance that, as representatives of a specific user group, they are assumed to possess expertise in certain functionalities. The aim is to observe how a user from a given group performs a task step by step while the system under test (SUT) is in use, with participants ‘verbalizing their actions aloud’ and describing the progression of the intended use-case workflow in their own words. When multiple users perform and verbalize the workflow simultaneously, a rapid overall situational understanding can be achieved. However, scheduling such group-based interview sessions requires substantial scheduling effort like making the bookings and so on and typically results in high costs in an industrial context. It is of course worth mentioning, how effective it is to implement a certain feature or workflow with the first try instead of continuous refactoring.

Sometimes unfortunately valuable silent data derived from individual observation and experiential learning may remain unnoticed if quieter participants have already identified more efficient workflows, but are unable to verbalize these insights during the interview sessions or workflows remain unnoticed by the researcher during data collection, analysis or reporting, as mistakes happen to all.

While ‘think-aloud’ techniques generate valid data efficiently and can be reliably verified through replication of a similar research setup, the elicitation of silent data is better supported through one-to-one interviews in my opinion. In cases where interviewees demonstrate substantially higher expertise, the extensive use of probing questions is recommended, as single questions are typically way too insufficient to surface *tacit* knowledge comprehensively. In short, interviews should employ iterative probing extensively to elicit detailed silent data, which constitutes a primary objective of the studies anyhow.

I have observed that when encountering highly experienced participants across different industry sectors, interviewers become increasingly aware of the value of deep interviewing, which often generates strong engagement and true enthusiasm on both sides. I experienced such effects firsthand while conducting research at Aalto University, Taloustutkimus Oy, JTL Jurionominen Tutkimuslaitos Oy and IRO Market Research Oy.

1.4.8. Transcriptions and synthetizing recommendations

Now the time has arrived to communicate the improvement proposals generated through heuristic evaluation, the chained cross-LLM prompt template, semi structured interviews and think aloud observations to the clients or own in-house development team. The findings produced through heuristic observation and the chained cross LLM prompt template may also be provided to the client's development team prior to the semi structured and think aloud directed interviews, as this may already improve the system's usability by a lot and therefore already support the interview process itself.

It has previously been identified which aspects should be reported using the applied template in order to enable the efficient definition of new requirement specifications within the development teams' backlogs. In **table 13** is one example presented how the recommended solution template could be structured visually informatively and how requirements may also be expressed using a User Story template, including possible acceptance criteria. However, I have paid attention to industry practice in recent years and observed that the usage of the backlog has been diminishing on too many occasions as sub tickets can easily be implemented even in CI/CD pipelines contexts with the AI Assistants in a matter of minutes. This is problematic even for the near future maintainability and I strongly believe that the requirements should always be traceable not just in the backlog files but also in the architectural documentations in relevant parts. Unfortunately, my recent observations from the industry especially those implementing the Agentic systems is such, that they consider the Scrum framework and backlog management historical events as the code can be generated, tested and documented within minutes but lets get back to the main issue of transcriptions.

In general, negative aspects should receive the primary focus in templates that document identified usability flaws. However, any final report delivered to the client at the end of the usability engineering evaluation process should also include positive aspects in order to ensure that already effective implementations remain unchanged. More importantly, the inclusion of positive findings creates an opportunity for both the external test consultant and the client, within their routine-driven working day, to re-evaluate whether successful implementations could be further improved even more. Additionally, this approach supports the maintenance of a rare but positive face-to-face discussion atmosphere when a typical extremely extensive list of critical findings related to identified weaknesses is being reviewed together.

The template below includes the following key viewpoints for rapid visualization:

Table 13. Usability engineering evaluation summary table for requirement proposals.

RESULTS: Finding #1	
User group:	
Use case:	
Location:	
Severity rating (1-4):	
Heuristic # 5: Error prevention	
Improvement suggestion.	
New User Story with acceptance criteria	
Screenshot	

1.5. Analysis

Now that the descriptive phase of the below activity diagram has been conducted by the literature review and experimentations, the lengthy workflow can be nicely synthesized as a coherent activity diagram illustrating the logic of the created design science artifact.

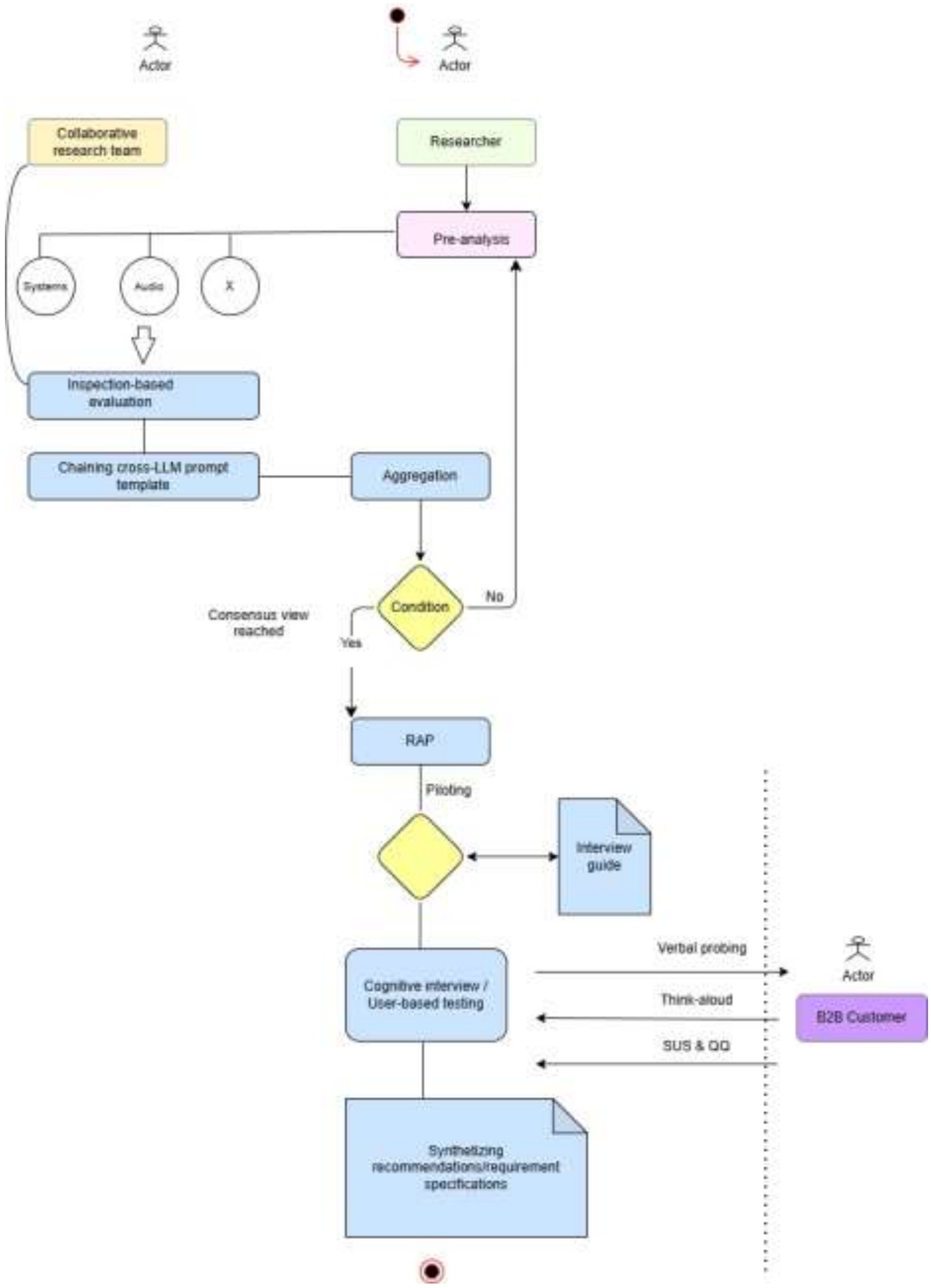


Figure 25. PUSA-X inspired activity diagram.

1.6. Discussion and conclusions

A pragmatic process-oriented PUSA-X design model for usability engineering process evaluation is proposed for complex systems. Instead of using some transdisciplinary discipline as a method of validation, the validation took place by using *transdisciplinary* idea as a core research methodology. The focus was to enhance utility by applying the best practices, as such tactics may improve the usability for the served stakeholders. Next, I will discuss the key observations by using the Hevner's summary table and Dudley-Evans (1986) model.

Table 14. Design-Science research guidelines.

Guideline	**Description**	**Alignment with RQ2:** How can an activity diagram be constructed as a rigorously grounded design science artifact that formalizes the essential stages of a collaborative usability engineering evaluation process and supports systematic utility improvement, thereby increasing usability in complex systems development contexts?
1: design as an artifact	Design-science research must produce a viable artifact in the form of **constructs** (vocabulary and symbols), **models** (abstractions and representations), **methods** (algorithms and practices) or an **instantiation** (implemented and prototype systems).	The novel design artifact is defined as an **abstracted model**: a novel UML *activity diagram* that formalizes the core logic steps of collaborative usability engineering process evaluation in support of development activities.
2: Problem relevance	The objective of design-science research is to develop technology-based solutions to important and relevant business problems.	The visualization of more rational steps in usability evaluation is intended to facilitate faster utility improvements.
3: Design evaluation	The utility, quality and efficacy of a design artifact must be rigorously demonstrated via well-executed evaluation methods.	The proposed design model visualized as an UML activity diagram is composed of independent analytical stages that are advanced logically toward subsequent phases and both the SUS and a novel QQ evaluation metric is instantly available to produce a quantitative outcome that facilitates further optimization for utility improvement purposes.
4: Research contributions	Effective design-science research must provide clear and verifiable contributions in the areas of the design artifact, design foundations and/or design methodologies.	The abstracted model is defined as an *activity diagram* in which the stages of usability evaluation are formalized into a logically progressing sequence and faster feedback is enabled with the use of the quantitative SUS and QQ metric, which are also readily verifiable by others.
5: Research rigor	Design-science research relies upon the application of rigorous methods in both the construction and evaluation of the design artifact.	The determination of the abstracted model is grounded in established frameworks from usability engineering and HCI (e.g., heuristic evaluation, usability testing methods, ISO standards, logic and UML), through which theoretical and methodological rigor is ensured.
6: Design as a search process	The search for an effective artifact requires utilizing available means to reach desired ends while satisfying laws in the problem environment.	An iterative design process was employed in which alternative activity diagram structures were explored and refined to identify an effective and generalizable usability evaluation model, which is then abstracted as an activity diagram.
7: Communication of research	Design-science related research must be presented effectively both to technology-oriented as well as management-orientated audiences.	Effective communication to both technology-oriented and management-oriented audiences is supported by the resulting activity diagram, which is used both as a formalized model and, for practitioners, both as an actionable guide for rapid usability evaluation and utility increasement visualization tool supported by a quantitative score deriving from the SUS and QQ metrics.

The activity diagram summarizes neatly the usability evaluation process and its close integration with the engineering process commonly described as the V-model. It can be argued that the usability evaluation is a type of usability testing which is part of the final acceptance testing process “to validate that the system serves the end-user requirements”.

Overall, the results are exciting and not what was expected in the beginning. The activity diagram became crowded due to the many required workflows on the hardware side and the addition of GenAI-based evaluation to support individual *inspection-based* evaluation.

In the past such activity diagrams had less logical steps to be modeled. The biggest problem I noticed also in this revised usability evaluation process was the amount of rational feedback. An individual conducting inspection-based evaluation can easily generate over fifty observations for a single screen view and the leading LLM's does the same unless explicitly told to keep it less. An average system has so many screen views that the final amount of rational feedback is simply too much to take into full consideration.

The working hypothesis was that it is possible to create a design science artifact from the usability evaluation process following an UML activity diagram logic and it was.

In the past it was a standard process to first conduct the usability evaluation so that one could determine how the new features might increase the end-user utility. I argue now instead that the additional utility can oftentimes be instantly added to the system with this novel PUSA-X design artifact. There are three main reasons for this:

i.) Involving more relevant engineering disciplines ensures that some critical success factors, such as, CPU utilization zones are safe enough, the device recharging times and special audio effects are not completely forgotten.

ii.) If prompted accordingly, the cross-LLM prompt template can provide recommendations from the world leading GenAI systems and typically avoid giving advice causing the overall usability to decrease.

iii.) Previous research suggests that the proper usability evaluation and sometimes even formulation of specific usability goals becomes well before implementing new utility increasing features. However, when seeing some totally forgotten functionalities in complex systems and getting rational feedback from the leading GenAI, then one can re-question the need for the usability evaluation before implementing additional utility increasing features. In the long run the lengthy usability evaluation process is of course required.

The pragmatic value of the artifact is there, and its usage can be measured with the SUS and QQ metric. One needs to apply the artefact in own organizational circumstances to see what other heuristics are also efficient, as Nielsen's heuristics are just a starter.

The methodological choice of key literature review together with experimentation with the cross-LLM prompt template produced new eye-opening results. In addition, most of the required qualitative research methods illustrated in the study are such that their mastery comes slowly only after some time in conducting interviews.

When I now evaluate my own research work, it is clear for me to see, that it would not have been possible if I did not have the previous long working experience in this area. The GenAI will affect the usability evaluation in the individual *inspection-based* evaluations phase also in the future. The *transcriptions* handling phase could also be delegated to some AI tools, but that area is left for future researchers to investigate further.

The activity diagram artefact illustrates that usability engineering process evaluation is largely collaborative and that the modeling of the process itself is grounded in the existing HCI theory base within the field of usability engineering (UE). The level of abstraction of the activity diagram is intended to capture the logical steps in a manner that is both generalizable, verifiable and actionable, such that the process results in measurable improvements in usability as assessed by the System Usability Scale (SUS) or QQ metric and consequently leads in increased utility. In conclusion, this overview chapter contributes to a pragmatic "what works", process-oriented design model modeling for usability engineering.